Indiranagar Exchange PE Node (IND-PE-03) - Hardware/Environmental Fault Failure (ASIC memory parity errors, line card crashes, power supply redundancy loss) - Critical Severity

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_7385

Description

On the Ericsson PE Node (IND-PE-03), located at the Indiranagar Exchange, hardware/environmental faults can occur due to the inherent vulnerability of the ASIC memory parity error detection mechanism. This can lead to a catastrophic failure of the line card, causing a complete network disconnection. Furthermore, the loss of power supply redundancy exacerbates the situation, resulting in an unmitigated hardware failure. The Line Card Crash and ASIC memory parity errors can be caused by various factors such as high temperatures, voltage fluctuations, and manufacturing defects. It is crucial to investigate the root cause to prevent a similar failure in the future.

Pre-requisite Checks

- Verify that the Indiranagar Exchange site is accessible and safe for remote hands to perform the necessary tasks.

- Confirm that the redundant link status is up and running to prevent any network disruptions.

- Ensure that the Ericsson CLI access is available and authenticated for the PE Node (IND-PE-03).

Potential Root Causes

Cause	Probability
ASIC memory parity error due to manufacturing defect	High
Line card crash due to high temperatures	Medium
Power supply redundancy loss due to voltage fluctuations	Low

Diagnostic Commands & Expected Outputs

```
Command: show controller line-card /all/asic-memory, Expected Output: ASIC memory parity error detected, Description: This command checks for any existing ASIC memory parity errors on the line card. A positive output indicates the presence of a parity error, which may be the root cause of the failure.

Command: show controller line-card /all/temperature, Expected Output: Line card temperature: 60°C, Description: This command verifies the current temperature of the line card. Elevated temperatures can cause the line card to crash, leading to a hardware failure.
```

```
Command: show controller power-supply /all/status, Expected Output: Power supply redundancy lost, Description: This command confirms the status of the power supply redundancy. A lost redundancy can lead to a complete hardware failure in case of a power supply failure.
```

Resolution Steps

Step 1: Step: Identify and replace faulty line card, Command: replace controller line-card, Description: This step involves identifying the faulty line card and replacing it with a new one. The `replace controller line-card` command initiates the replacement process.

Step 2: Step: Verify power supply redundancy, Command: verify controller power-supply /all/status, Description: This step confirms that the power supply redundancy is restored after replacing the line card. The `verify controller power-supply /all/status` command checks the status of the power supply redundancy.

Step 3: Step: Clear ASIC memory parity errors, Command: clear controller line-card /all/asic-memory, Description: This step clears any existing ASIC memory parity errors on the line card. The `clear controller line-card /all/asic-memory` command resets the ASIC memory parity error counter.

Post-Check Verification

```
Command: show controller line-card /all/asic-memory, Expected Output: ASIC memory parity error cleared, Description: This command verifies that the ASIC memory parity error has been cleared after executing the `clear controller line-card /all/asic-memory` command.

Command: show controller line-card /all/temperature, Expected Output: Line card temperature: 30°C, Description: This command confirms that the line card temperature has returned to a normal range after replacing the faulty line card.

Command: show controller power-supply /all/status, Expected Output: Power supply redundancy restored, Description: This command verifies that the power supply redundancy has been restored after replacing the line card.
```

Roll-back Procedures

- Step: Restore original line card, Command: restore controller line-card, Description: In case the replacement line card does not resolve the issue, this step involves restoring the original line card to its previous state. The `restore controller line-card` command reverses the replacement process.

- Step: Re-enable power supply redundancy, Command: enable controller power-supply /all/redundancy, Description: If the power supply redundancy was disabled during the resolution process, this step re-enables it. The `enable controller power-supply /all/redundancy` command restores the power supply redundancy.

- Step: Clear ASIC memory parity errors (again), Command: clear controller line-card /all/asic-memory, Description: If the ASIC memory parity error was not cleared during the resolution process, this step clears it again using the `clear controller line-card /all/asic-memory` command.